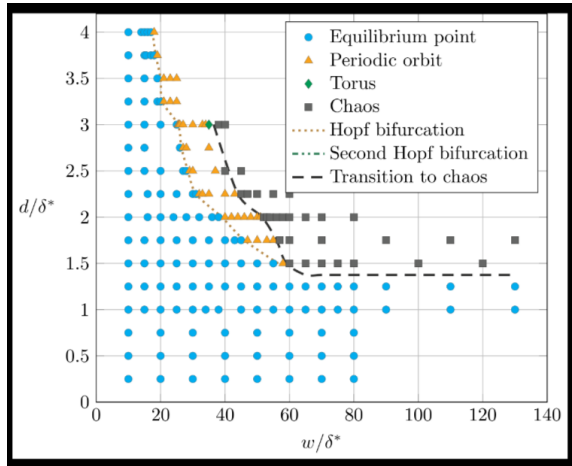


IMPERIAL

Torus diagram + SPOD

Víctor Ballester
October 22, 2025

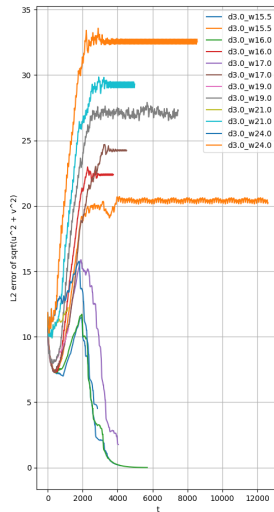
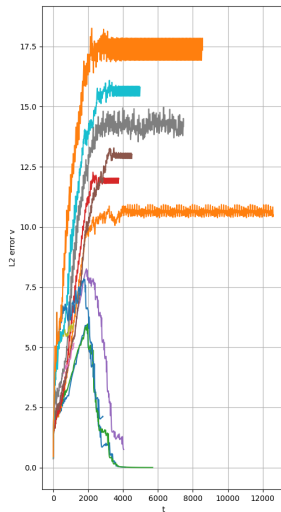
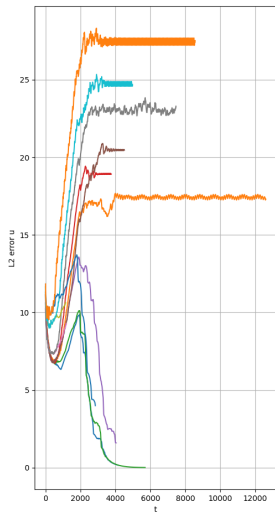
Reminder



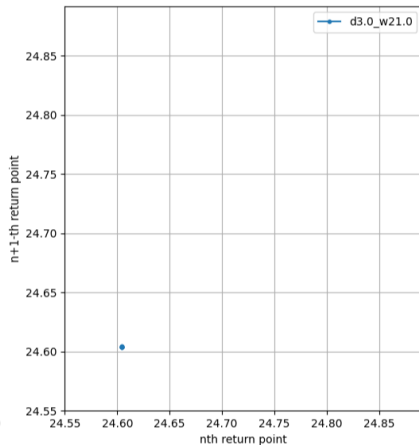
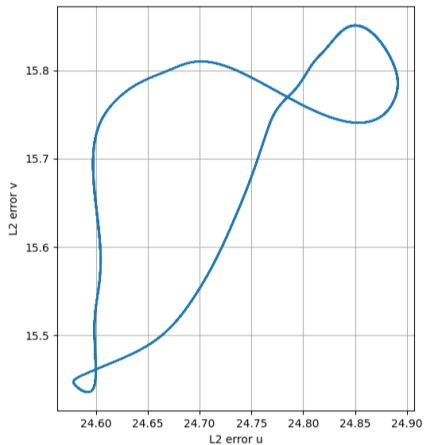
Edge state analysis for $d = 3$

- As we approach the Hopf bifurcation at $w \sim 25.4$, the energy of this other periodic orbit increases, so I think those two states are unrelated.
- The L2 energy of the periodic orbit at $w = 26$ is very low (of course, because it is only localized near the gap) around 0.47.
- Also I think there are many more weird attractors without a clear pattern than just the ones drawn in the previous diagram. because for some reason $w = 16$ behaves differently than the other cases.

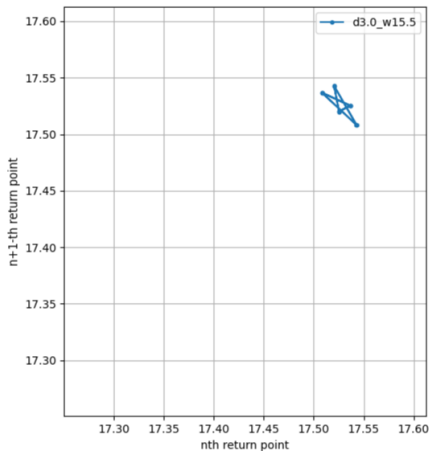
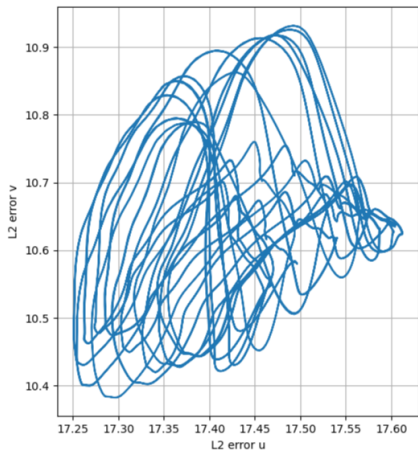
Edge state energy



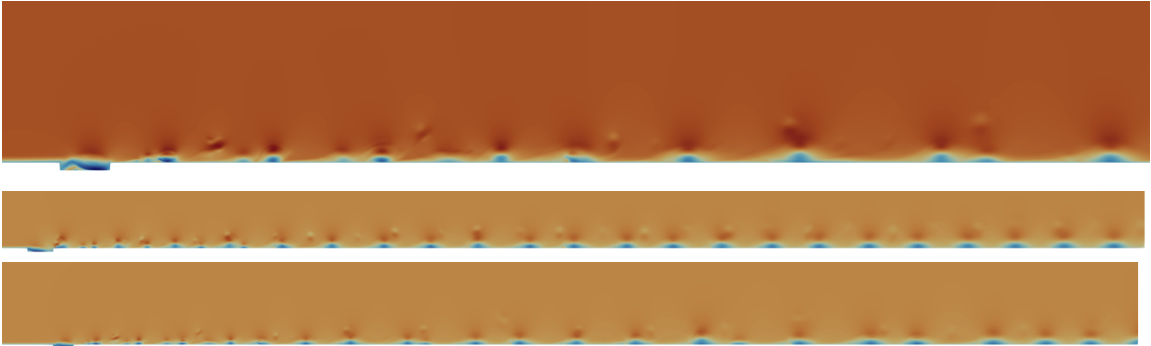
P.O. full BL excited at $d = 3$ and $w = 21$



quasi-periodic? full BL excited at $d = 3$ and $w = 15.5$

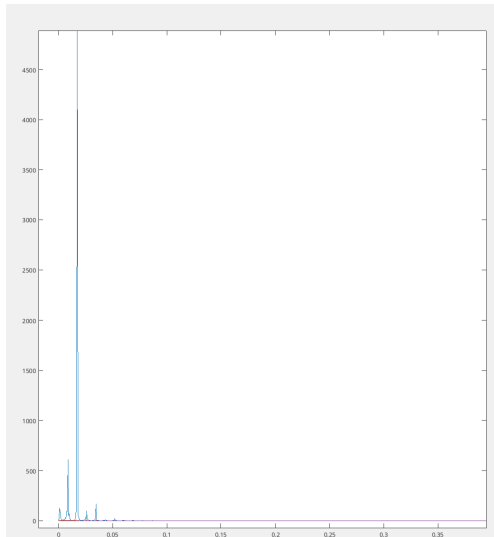


u fields

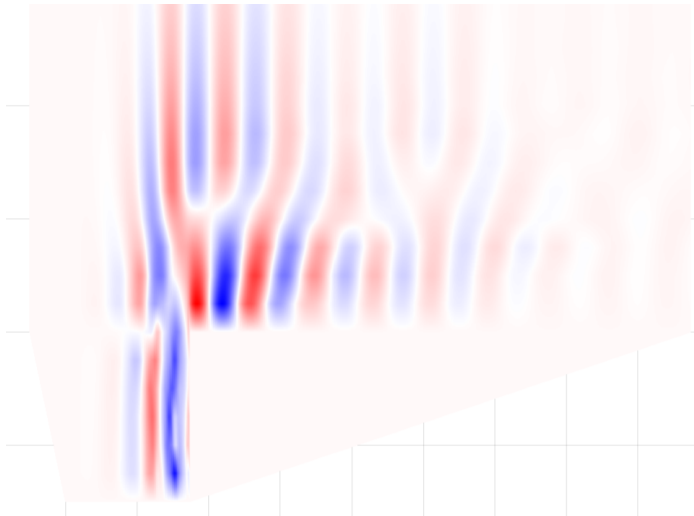


SPOD analysis

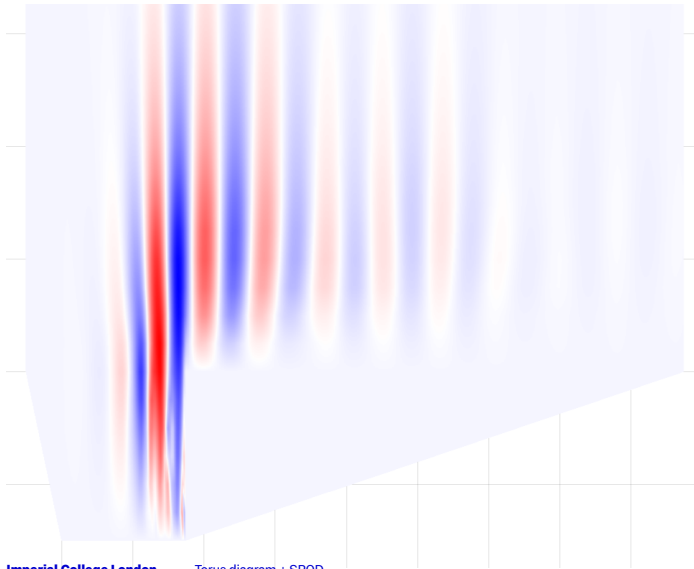
- Main mode: $A = 5170, \omega = 0.107$.
- Second mode: $A = 620, \omega = 0.055$.



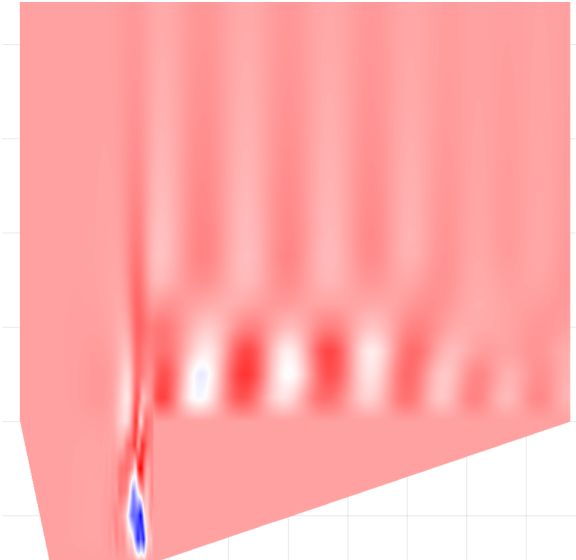
SPOD 1st mode u



SPOD 1st mode v



SPOD 2nd mode u



SPOD 2nd mode v

